Does Doubling the Halfway Split Predict the Finish? A Simple Linear Regression of the 2023 Boston Marathon

Pujitha Bobbili

October 30, 2026

The code used to download, prepare, and analyze the data is available at https://github.com/pujithabobbili/simple-regression-project.

Abstract

Doubling a runner's halfway time is a common way to project a marathon finish, but it assumes that the two halves take equal time. Using official results for 26,526 finishers of the 2023 Boston Marathon (Fay et al. 2023), I fitted a simple linear regression of net finishing time on half-marathon split, both in minutes, and compared the slope with 2, the value implied by equal halves. The estimated slope was 2.218 (95% confidence interval 2.210 to 2.225), so each extra minute at halfway is associated with about 1.22 extra minutes in the second half. Doubling therefore underestimates the average finish by about 5 minutes at a 1:30 split and about 18 minutes at a 2:30 split. Because finishing times also spread out more for slower runners, a 95% prediction interval for one runner is about 25 minutes wide at a 1:30 split but about 54 minutes wide at 2:30, so the fitted line gives a good estimate of the average finish but a less certain one for any single runner, especially a slower one.

1 Introduction

If a runner reaches the halfway point of a marathon in 1 hour 45 minutes, doubling that time gives a projected finish of 3 hours 30 minutes. This works exactly only when the runner completes the second half in another 1 hour 45 minutes. Any slowdown after halfway makes the projection too optimistic.

Running the second half more slowly than the first is called a positive split. Previous studies have found that positive splits are common in marathons. March et al. (2011) and Santos-Lozano et al. (2014) reported differences in pacing across runners of different abilities, and March et al. found that slower and less experienced runners tended to slow more. Earlier research also found that men slowed more than women on average (Deaner et al. 2015), that warmer conditions were associated with larger slowdowns, especially among slower runners (Ely et al. 2008), and that some recreational runners have severe late-race slowdowns (Smyth 2021). These studies compare groups of runners, but they do not give one finishing-time equation that a runner could use at halfway.

In this paper, I ask two questions. First, how closely does the halfway split predict net finishing time in the 2023 Boston Marathon? Second, how far does the doubling rule miss, and for whom? If every runner kept an even pace, finishing time would rise by exactly 2 minutes for each extra minute at halfway. Because positive splits are common and slower runners tend to slow more, I expect the slope to be above 2, so that doubling underestimates the finish, and by more for slower runners. To check this, I fitted a simple linear regression of net finishing time on half-marathon split for the 26,526 finishers with a recorded split.

The estimated slope is 2.218, so the expectation holds: an extra minute at halfway is associated with about 2.22 extra minutes at the finish, including about 1.22 extra minutes in the second half. The uncertainty of a

projection for one runner also grows with the split, so the halfway clock is less informative for slower runners. This matters because runners and spectators use the doubling rule to plan a race, and the rule is furthest off for exactly these runners. Section 2 describes the data, Section 3 explains the model and its checks, Section 4 presents the results, and Section 5 discusses their limits and possible extensions.

2 Data

2.1 Source and observational units

I used the official results of the Boston Marathon held on April 17, 2023. Fay et al. (2023) compiled the file from results published by the Boston Athletic Association (2023) and made it available through the SCORE Network Sports Data Repository. Each row represents one finisher. The full file has 26,598 finishers, so it covers the finishing field rather than a random sample of it. Runners who started but did not finish are not included. The times come from electronic timing chips, which record when a runner crosses mats along the course; they do not show what happened between the mats.

Since the complete finishing field is observed, the fitted line describes the runners in this race directly. The confidence intervals require a broader interpretation: I use them as model-based measures of uncertainty for similar runners racing under similar conditions, rather than as uncertainty from sampling the 2023 finishing field. Boston also has a selective entry process: most entrants meet a qualifying-time standard at an earlier marathon, and others enter through charity programs. Its runners are therefore not a random sample of all marathon runners, which limits how widely the results apply.

2.2 Variables

The source file contains 15 columns with runner information, finishing places, and race times. **Net finishing time**, the response, is the elapsed time between crossing the starting mat and crossing the finishing mat, converted from seconds to minutes. I used it rather than gun time, which starts at the official signal and so also includes the wait to reach the start line: in this file, gun time exceeds net time by a median of 2.5 minutes and by as much as 27.1 minutes. **Half-marathon split**, the explanatory variable, is the net elapsed time at the halfway mat, also in minutes. From these two, I calculated the second-half time (finishing time minus split) and the split ratio (second-half time divided by split); a ratio above 1 means a slower second half. Sex (42.9% women) and age group (ten bands, with 61.8% of finishers aged 40 or over) are used only to describe the field.

The times are recorded to about one second, which is negligible next to the 19.3-minute standard deviation of halfway splits. A more important point is that finishing time includes the halfway split, so a strong association between the two is built in.

2.3 Missing data and field characteristics

There are 72 finishers (0.27% of the field) without a recorded half-marathon split, probably because of a missed chip reading. I excluded them, leaving 26,526 runners. Their mean finishing time (224.4 minutes) is close to that of the runners retained (222.5 minutes), and their sex and age-group make-up is similar, so the exclusion is unlikely to matter. No other variable used here has missing values, and there are no duplicate runners.

Table 1 summarizes the analysis variables, and Figure 1 shows their distributions. All three times have long upper tails. The median split ratio is 1.058, so the typical finisher took about 5.8% longer for the second half, and 89.0% of finishers ran a positive split. Pacing was similar for women and men: both groups had a median split ratio of 1.058, even though women's median finish (3:46:04) was later than men's (3:20:56). All runners also faced the same course on the same day, so this file cannot separate weather from pacing; earlier work links heat to larger slowdowns, particularly for slower runners (Ely et al. 2008).

Table 1: Summary statistics for the 26,526 finishers with a recorded halfway split. Times are in minutes; the split ratio is second-half time divided by halfway split. All three times are right skewed, and the median split ratio of 1.058 means that the typical finisher ran the second half about 6% slower than the first.

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Half-marathon split	106.9	19.3	62.3	92.5	103.5	116.9	207.2
Net finishing time	222.5	43.7	125.9	191.1	213.9	244.4	440.5
Second-half time	115.6	25.3	63.6	97.7	110.1	127.9	248.4
Split ratio	1.078	0.086	0.526	1.022	1.058	1.113	2.524

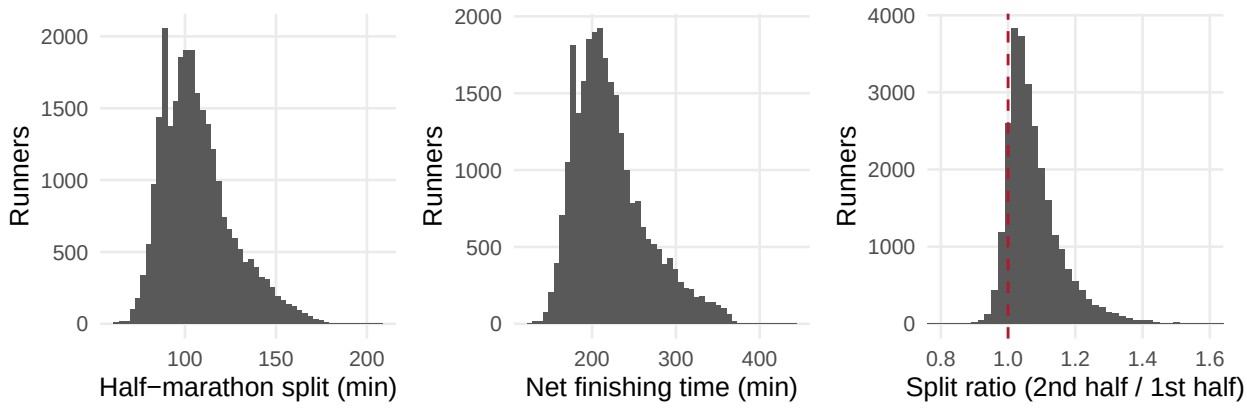


Figure 1: Distributions of halfway split, net finishing time, and split ratio. The dashed line in the right panel marks equal times for the two halves; its horizontal axis is limited to 0.8–1.6, which leaves out 0.15% of runners. Both times have long right tails, and most split ratios are just above 1, so most runners slowed in the second half.

2.4 Additional and alternative data

Several additional data sources could help explain the pacing pattern. The Boston Athletic Association publishes 5 km splits, which could show where a runner began to slow. Results from other years and other marathons would allow comparisons across courses and conditions. Hourly weather records from the National Oceanic and Atmospheric Administration could be matched to the period when runners were on the course. Qualifying times would provide information about recent performance before race day.

3 Methods

3.1 The model

For runner i, let Y_i be net finishing time and x_i be the half-marathon split, both in minutes, for $i = 1, \dots, n$ with $n = 26,526$. The simple linear regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

The intercept β_0 is the expected finishing time at a split of zero, which is impossible in a race, so it only positions the line. The slope β_1 is the expected difference in finishing time for a one-minute difference in split. The error ε_i is the difference between runner i's finishing time and the mean for runners with the same split; the errors are assumed independent and normal with mean 0 and constant variance σ^2. I estimated β_0 and β_1 by ordinary least squares, which minimizes the sum of squared vertical distances $\sum_{i=1}^{n}(Y_i - \beta_0 - \beta_1 x_i)^2$; the resulting estimates are written $\hat{\beta}_0$ and $\hat{\beta}_1$. I estimated σ by the residual standard deviation $\hat{\sigma}$.

If both halves took the same time for every runner, then $Y_i = 2x_i$, which corresponds to $\beta_0 = 0$ and $\beta_1 = 2$. A slope of 2 alone would not establish equal pacing, because a nonzero intercept could still produce a systematic difference between the halves, so I also compare the whole fitted line with the doubling rule. A slope above 2 means that runners who reach halfway later also lose more time in the second half. I answer the second research question with the 95% confidence interval for β_1, which shows how far the slope is from 2, and by comparing the fitted line with the doubling rule at several splits. I used one predictor because the question is about what can be estimated from the halfway clock alone, and I kept both variables in minutes so that the slope can be compared directly with 2.

For a runner with split x_0, the fitted finishing time is $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. A confidence interval describes the average finish of runners with that split, while a prediction interval describes one runner and must also allow for that runner's own distance from the line, so it is much wider.

3.2 Assumptions and diagnostics

Figure 2 shows the checks of the model assumptions.

Linearity. The residual plot shows no clear curve, so a straight line is a reasonable summary of the average relationship over the observed range.

Independence. The model treats the errors of different runners as unrelated. This is hard to justify fully in a race, because runners may follow the same pacing group or respond similarly to course conditions. If so, the data hold less independent information than 26,526 runners suggest, and even the robust standard error in Section 3.3 would be too small, making the slope's confidence interval too narrow. The slope estimate itself would not be biased, and the conclusion would not change: the slope is about 56 standard errors above 2, so its standard error would have to be about 29 times larger before the interval reached 2. The prediction intervals describe how far individual runners fall from the line, which dependence between runners changes little.

Constant variance. This assumption does not hold. The band of residuals widens as fitted times increase: the residual standard deviation rises from 5.9 minutes in the fastest tenth of the field to 14.1 minutes in the slowest, about a 2.4-fold increase. Section 3.3 explains how I handle this.

Normality. The histogram of residuals is centered near zero but right skewed, with a long upper tail of runners who finished much later than predicted. With this many runners, the confidence interval for the slope is still reliable, but a prediction interval that relies on normal errors is not.

I also examined Cook's distance, which measures how much the fitted line would move if one runner were removed. The largest value is 0.060, well below the usual reference value of 1, so no single runner drives the fit. I kept runners with unusual pacing because they are real race results.

3.3 Model selection and validation

I kept the model in its simplest form. Both variables stay in minutes, so the slope can be compared directly with 2, and the residual plot shows no clear curve, so a quadratic term was not needed. The failure of constant variance has two consequences. First, the usual standard errors assume one common variance. I therefore computed the standard errors and confidence intervals with a heteroskedasticity-consistent (HC1) estimator (Zeileis 2004), which lets each runner's residual contribute its own variance. This changes the intervals but not the fitted line.

Second, the usual prediction interval uses the single value $\hat{\sigma}$ for every runner, so it has nearly the same width at every split. To see whether that is a problem, I counted how often the usual 95% prediction interval contains the actual finishing time within each tenth of the field. It contains 98.4% of the fastest tenth but only 84.5% of the slowest, so it is too wide for fast runners and too narrow for slow ones. I therefore built prediction intervals from nearby runners instead. For a split x_0, I took the runners whose splits lie within 5 minutes of x_0 and added the 2.5th and 97.5th percentiles of their residuals to $\hat{Y}_0$. These intervals widen where the spread is larger and can be longer on the slow side, which matches the right skew of the residuals. They leave out the uncertainty in the fitted line itself, which is well under a minute at these splits and small next to the spread of runners.

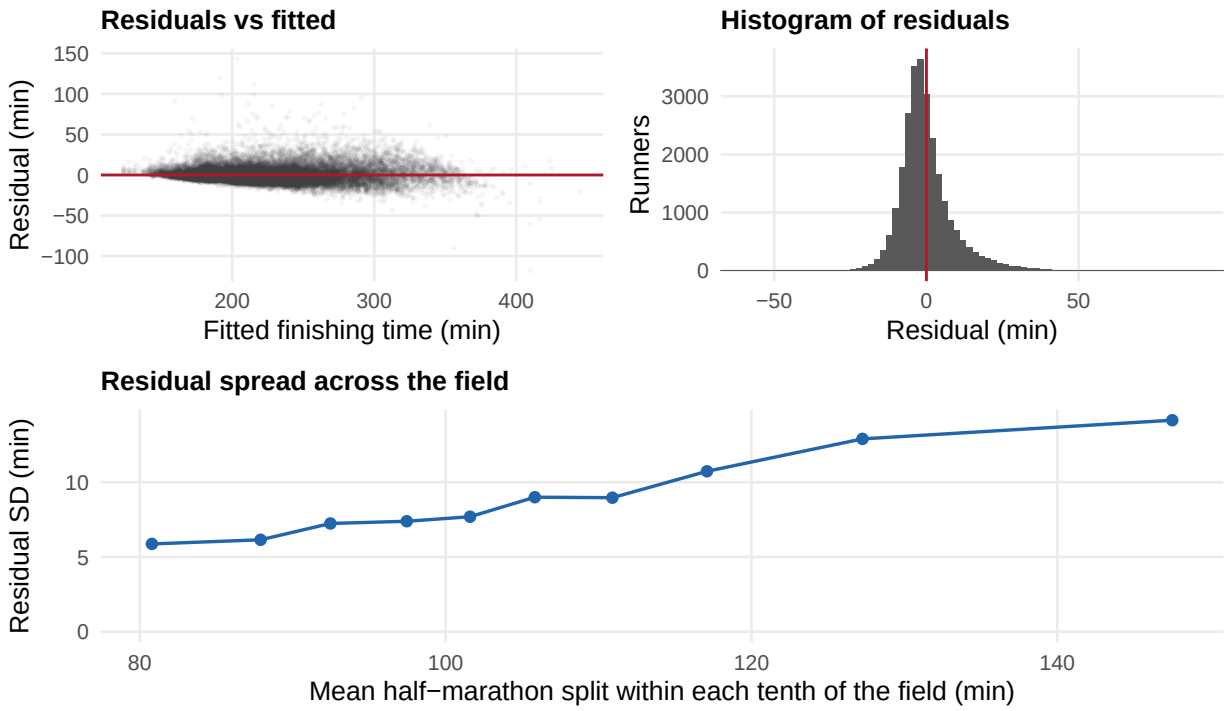


Figure 2: Regression diagnostics. Top left: residuals against fitted values show no clear curve but widen as fitted times increase. Top right: the histogram of residuals is centered near zero but right skewed, with a long upper tail; its horizontal axis is limited to −60 to 90 minutes. Bottom: the residual standard deviation within each tenth of the halfway split rises from about 6 to 14 minutes, so the constant-variance assumption does not hold.

3.4 Possible extensions to the model

Adding sex and age group, with interactions with the split, would let the slope differ between groups. Because women and men had similar split ratios, this would probably change the slope little, and it would need more information than the halfway clock provides. A quadratic term would let the line bend, but the residual plot shows no clear curvature. Weighted least squares or quantile regression would model the changing spread directly and could replace the nearby-runner intervals with a single model.

3.5 Software

The analysis was carried out in R version 4.6.1 (R Core Team 2026). I used `dplyr` for data handling (Wickham et al. 2023) and `readr` to read the data (Wickham et al. 2024). Figures were made with `ggplot2` (Wickham 2016) and combined using `patchwork` (Pedersen 2024). Robust standard errors came from `sandwich` (Zeileis 2004; Zeileis et al. 2020). The report used Quarto (Allaire et al. 2022), `knitr` (Xie 2023), and `kableExtra` (Zhu 2024). The regression was fitted with `lm` in R's `stats` package.

4 Results

4.1 The fitted model

The fitted line, with coefficients rounded for display, is

$$\widehat{\text{finish}} = -14.53 + 2.218 \times \text{half-marathon split}.$$

Both times are in minutes, and Table 2 gives the estimates.

Table 2: Least squares estimates for net finishing time regressed on halfway split, with heteroskedasticity-consistent (HC1) standard errors and 95% confidence intervals. The slope's interval, 2.210 to 2.225, lies entirely above 2, the value implied by equal halves.

Term	Estimate	Std. error (HC1)	95% CI
Intercept	-14.526	0.390	-15.291 to -13.762
Half-marathon split	2.218	0.00387	2.210 to 2.225

The slope's 95% confidence interval, 2.210–2.225, lies entirely above 2, so the fitted relationship is steeper than the doubling rule. For runners whose splits differ by one minute, the model estimates a difference of about 2.22 minutes at the finish, about 1.22 minutes of which comes in the second half. The negative intercept has no race meaning, since a zero-minute split is far outside the observed range of 62.3–207.2 minutes. The model has $R^2 = 0.953$, but much of this comes from finishing time containing the split, so the residual standard deviation of 9.43 minutes is a more useful measure of fit.

Figure 3 shows the fitted line with the doubling rule. The two lines cross at a split of about 1:06:41; only 30 runners were faster than that. For everyone else, the model predicts a later finish than doubling, and the gap grows with the split.

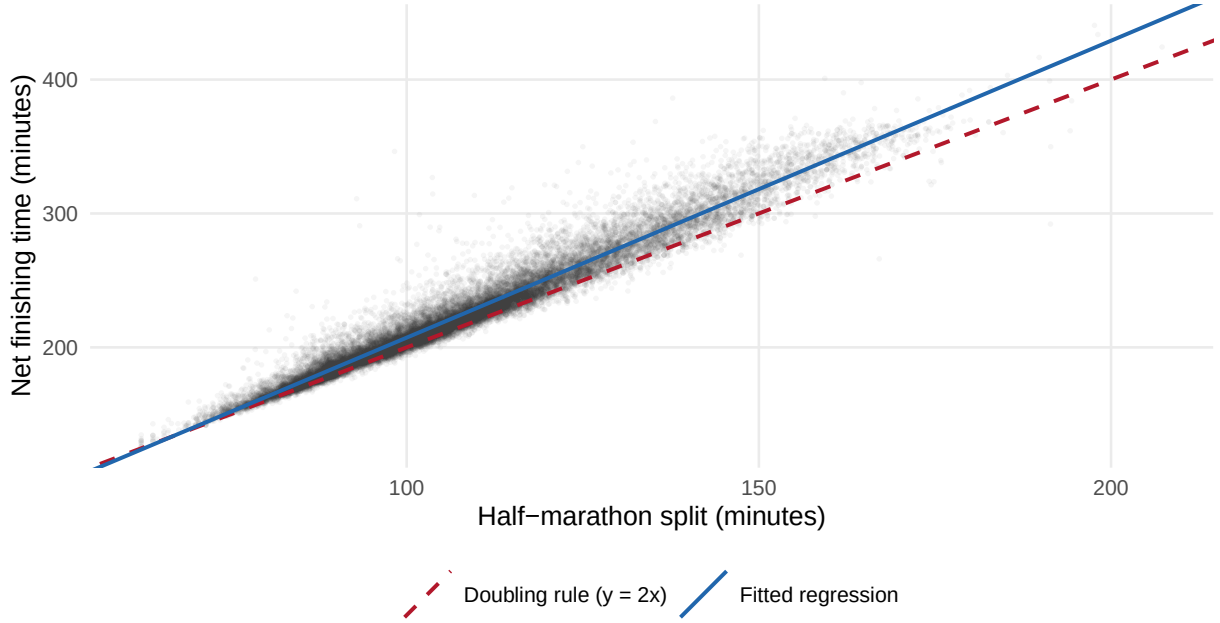


Figure 3: Net finishing time against halfway split for 26,526 finishers, with the fitted regression line (solid) and the doubling rule, finish $= 2 \times$ split (dashed). Darker areas hold more runners. The lines cross near a 1:07 split; above that, the fitted line lies above the doubling line, and the gap grows for slower runners.

4.2 What the model means for a runner

Table 3 gives projected finishing times for five halfway splits. At a 1:30 split, the model adds about 5 minutes to the doubled split; at 2:30, it adds about 18 minutes. For the median split of 1:43:33, the difference is about 8 minutes. A rough mental version is to double the split and add one minute for every five minutes beyond 1:07, since $\widehat{\text{finish}} - 2x = -14.53 + 0.218x$.

Table 3: Finishing-time projections for five halfway splits. 'Doubling short by' is the model projection minus the doubled split. Each 95% prediction interval adds the 2.5th and 97.5th percentiles of the residuals of runners within 5 minutes of that split to the model projection, so it widens for slower splits and is longer on the slow side.

Halfway	Doubling	Model	Doubling short by	95% prediction interval	Width
1:30:00	3:00:00	3:05:05	5.1 min	2:57:47 to 3:23:14	25 min
1:45:00	3:30:00	3:38:21	8.3 min	3:26:47 to 3:58:47	32 min
2:00:00	4:00:00	4:11:37	11.6 min	3:55:46 to 4:39:41	44 min
2:15:00	4:30:00	4:44:53	14.9 min	4:25:14 to 5:18:13	53 min
2:30:00	5:00:00	5:18:09	18.1 min	4:56:04 to 5:50:22	54 min

The prediction intervals show how uncertain a projection is for one runner. They are about 25 minutes wide at a 1:30 split and about 54 minutes wide at 2:30, compared with about 37 minutes at every split for the usual interval. They are also uneven: at a 1:45 split, a runner may finish up to about 12 minutes faster than projected but up to about 20 minutes slower, because some runners slow down sharply in the second half.

5 Discussion

For the 2023 Boston finishers, the halfway split predicts the finish closely, but not in the way the doubling rule assumes. The fitted slope is 2.218, and its confidence interval lies above 2, as expected from how common positive splits are: 89% of runners ran a slower second half. Doubling therefore underestimates the average finish by about 5 minutes for a 1:30 split and about 18 minutes for a 2:30 split. This agrees with earlier findings that slower runners slow more (March et al. 2011; Santos-Lozano et al. 2014), and it summarizes the pattern in one equation that a runner can use at halfway.

The average finish is estimated very precisely, but one runner's finish is not. The 95% prediction interval is about 25 minutes wide for a 1:30 split and about 54 minutes wide for a 2:30 split, so a projection should be read as an estimate of the mean, with more uncertainty for slower runners.

The study has several limitations. The data include finishers only, so runners who dropped out, possibly after slowing sharply, are missing, and the results do not describe all starters. Boston's field is selective and raced one course on one day, so a different race could produce a different slope. Runners who race together may have related errors, which the model ignores. The nearby-runner intervals are built and checked on the same data, so they should be tested on another race before being used as forecasts. Finally, the model is descriptive: it does not show that starting faster would make a runner finish sooner.

Several extensions would be useful. The organizers' 5 km splits would show where the slowdown begins, which relates to the late-race collapse studied by Smyth (2021). Fitting several years with weather data would show whether the slope changes with temperature (Ely et al. 2008), and adding sex and age group could test the differences between groups reported by Deaner et al. (2015).

References

Allaire, J. J., Charles Teague, Yihui Xie, and Christophe Dervieux. 2022. *Quarto*. https://doi.org/10.5281/zenodo.5960048.

Boston Athletic Association. 2023. *2023 Boston Marathon Official Results*. Boston Athletic Association. https://www.baa.org.

Deaner, Robert O., Rickey E. Carter, Michael J. Joyner, and Sandra K. Hunter. 2015. "Men Are More Likely Than Women to Slow in the Marathon." *Medicine & Science in Sports & Exercise* 47 (3): 607–16. https://doi.org/10.1249/MSS.0000000000000432.

Ely, Matthew R., David E. Martin, Samuel N. Cheuvront, and Scott J. Montain. 2008. "Effect of Ambient Temperature on Marathon Pacing Is Dependent on Runner Ability." *Medicine & Science in Sports & Exercise* 40 (9): 1675–80. https://doi.org/10.1249/MSS.0b013e3181788da9.

Fay, Jack, A. J. Dykstra, and Ivan Ramler. 2023. *2023 Boston Marathon Runners*. SCORE Network Sports Data Repository. https://data.scorenetwork.org/running/boston_marathon_2023.html.

March, Daniel S., Paul M. Vanderburgh, Peter J. Titlebaum, and Meghan L. Hoops. 2011. "Age, Sex, and Finish Time as Determinants of Pacing in the Marathon." *Journal of Strength and Conditioning Research* 25 (2): 386–91. https://doi.org/10.1519/JSC.0b013e3181bffd0f.

Pedersen, Thomas Lin. 2024. *patchwork: The Composer of Plots*. https://patchwork.data-imaginist.com.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

Santos-Lozano, Alejandro, Pilar S. Collado, Carl Foster, Alejandro Lucia, and Nuria Garatachea. 2014. "Influence of Sex and Level on Marathon Pacing Strategy. Insights from the New York City Race." *International Journal of Sports Medicine* 35 (11): 933–38. https://doi.org/10.1055/s-0034-1367048.

Smyth, Barry. 2021. "How Recreational Marathon Runners Hit the Wall: A Large-Scale Data Analysis of Late-Race Pacing Collapse in the Marathon." *PLoS ONE* 16 (5): e0251513. https://doi.org/10.1371/journal.pone.0251513.

Wickham, Hadley. 2016. *ggplot2: Elegant Graphics for Data Analysis*. 2nd ed. Springer-Verlag New York. https://ggplot2.tidyverse.org.

Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2023. *dplyr: A Grammar of Data Manipulation*. https://dplyr.tidyverse.org.

Wickham, Hadley, Jim Hester, and Jennifer Bryan. 2024. *readr: Read Rectangular Text Data*. https://readr.tidyverse.org.

Xie, Yihui. 2023. *knitr: A General-Purpose Package for Dynamic Report Generation in R*. https://yihui.org/knitr/.

Zeileis, Achim. 2004. "Econometric Computing with HC and HAC Covariance Matrix Estimators." *Journal of Statistical Software* 11 (10): 1–17. https://doi.org/10.18637/jss.v011.i10.

Zeileis, Achim, Susanne Köll, and Nathaniel Graham. 2020. "Various Versatile Variances: An Object-Oriented Implementation of Clustered Covariances in R." *Journal of Statistical Software* 95 (1): 1–36. https://doi.org/10.18637/jss.v095.i01.

Zhu, Hao. 2024. *kableExtra: Construct Complex Table with 'kable' and Pipe Syntax*. https://haozhu233.github.io/kableExtra/.